

The Social Value of Financial Privacy

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1 A simple model of financial privacy

1.1 Setup

Consider a discrete-time economy with a single token worth v units of numéraire. The token is traded by $n \geq 2$ investors. Investors are of two types: they either hold a high token inventory $I = H$ with probability α , or a low token inventory $I = L$ with probability $1 - \alpha$, where $H \geq L > 0$ and $2L > \mathbb{E}I$.

Holding inventory is costly. If an investor ends the game with cash w_i and inventory I_i , her terminal utility is

$$U_i(w_i, I_i) = w_i + vI_i - \frac{\gamma}{2}I_i^2, \quad (1)$$

where $\gamma > 0$ measures aversion to inventory risk. The quadratic inventory penalty can be interpreted as a need to unwind the position. In practice, it would be driven by liquidity needs, risk management requirements, or regulatory constraints.

Investors can unwind inventory by selling tokens to a monopolist risk-neutral *dealer*. A trader with inventory I_i who sells $q \in [0, I_i]$ at price p obtains utility

$$U_i(q | I_i) = (I_i - q)v + pq - \frac{\gamma}{2}(I_i - q)^2. \quad (2)$$

The first term $(I - q)v$ represents the fundamental value of the retained inventory. The second term pq is the revenue from selling q units at price p . The third term captures the convex cost of holding any residual inventory.

1.2 Timing

The game proceeds as follows. At $t = 0$, investors may bid for a private wallet that hides token balances from the dealer. Unsuccessful bidders receive a public wallet, allowing the dealer to condition the purchase price on the investor token balance.

At $t = 1$, each investor receives either H or L tokens in her chosen wallet and may subsequently reveal her inventory to the dealer. Next, at $t = 2$, investors submit requests for quote (RFQs) to sell

tokens. The dealer responds with a uniform price $\hat{p}_i$ per RFQ: that is, he stands ready to buy any amount at this price from investor i . Quotes may depend on inventory when holdings are public; otherwise the dealer posts a pooling price for private wallets.

After observing the quote, each investor selects a trade quantity to sell, that is $q \in [0, I_i]$ at $t = 3$. Finally, payoffs are realized at $t = 4$. Figure 1 illustrates the timeline.

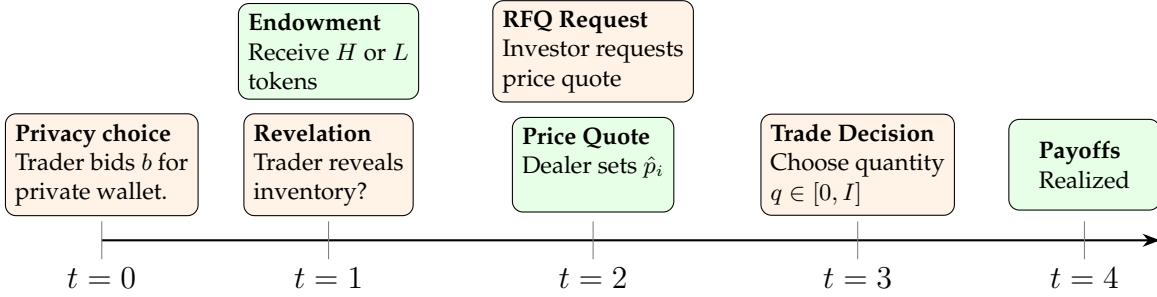


Figure 1: Timeline of the trading game

Two assumptions are key. First, privacy is acquired *before* inventories are realized, so the dealer cannot infer types from the initial privacy decision. Second, RFQs omit quantities, so order size does not reveal type. Thus the dealer can learn investor type only from explicit revelation at $t = 1$.

1.3 The insurance value of privacy

To build intuition for the insurance role of privacy, we begin with a simplified setting in which traders cannot reveal inventories at $t = 1$; that is, privacy choices made at $t = 0$ are binding and irreversible.

We solve the game by backward induction. At $t = 3$, an investor facing price $\hat{p}$ chooses q to maximize (2), yielding the optimal liquidation quantity:

$$q^*(I, p) = \max \left\{ 0, I - \frac{v - \hat{p}}{\gamma} \right\}. \quad (3)$$

Trade volume increases with larger initial inventory I , higher inventory aversion γ , or better dealer prices $\hat{p}$. At $t = 2$, the optimal dealer pricing depends on whether the request for quote comes from a public or private wallet. When inventories are public, the dealer observes I and sets p to

maximize profit:

$$\max_p \Pi(p|I) = (v - p) \cdot q^*(I, p) = (v - p) \left(I - \frac{v - p}{\gamma} \right), \quad (4)$$

yielding the optimal public wallet price

$$p_{\text{public}}^*(I) = v - \frac{\gamma I}{2}. \quad (5)$$

Dealers extract surplus by offering discounts below the fundamental value v . Higher inventory investors face worse prices, enabling perfect price discrimination. From equations (3) and (5), an investor with a public wallet optimally sells half their inventory:

$$q_{\text{public}}(I) = I - \frac{v - p_{\text{public}}^*(I)}{\gamma} = I - \frac{\gamma I/2}{\gamma} = \frac{I}{2}. \quad (6)$$

With private wallets the dealer cannot condition on I and instead maximizes expected profit,

$$\begin{aligned} \max_p \Pi(p) &= \mathbb{P}(L) (v - p) \cdot q^*(L, p) + \mathbb{P}(H) (v - p) \cdot q^*(H, p) \\ &= (v - p) \left(\mathbb{E}[I] - \frac{v - p}{\gamma} \right), \end{aligned} \quad (7)$$

where $\mathbb{E}[I]$ is the expected inventory, which delivers the pooling price

$$p_{\text{private}}^* = v - \frac{\gamma \mathbb{E}[I]}{2}. \quad (8)$$

All private investors face the same quote. Given (8), type- I investor sells $q_{\text{private}}(I) = I - \frac{\mathbb{E}[I]}{2}$. High-type investors sell more than half of their inventory; low-types sell less. Privacy therefore induces cross-subsidization as high types benefit at the expense of low types.

Proposition 1. [Privacy value as a hedge] *For investors, the insurance value of purchasing a private wallet is equal to*

$$\nu_{\text{binding}} = \mathbb{E} \left[U_{\text{private}}^{\text{eq}} \right] - \mathbb{E} \left[U_{\text{public}}^{\text{eq}} \right] = \frac{3\gamma}{8} \left(\mathbb{E}[I^2] - \mathbb{E}[I]^2 \right) = \frac{3\gamma}{8} \text{var}(I), \quad (9)$$

which equals the investor's maximum willingness to pay.

Proof. To obtain the result in Proposition 1, we note that for a public investor, the expected utility is

$$\mathbb{E}U_{\text{public}}^{\text{eq}} = \mathbb{E} \left[\frac{I}{2} \cdot v + \left(v - \frac{\gamma I}{2} \right) \cdot \frac{I}{2} - \frac{\gamma}{2} \left(\frac{I}{2} \right)^2 \right] = \mathbb{E} \left[vI - \frac{3\gamma I^2}{8} \right] \quad (10)$$

$$= v\mathbb{E}[I] - \frac{3\gamma}{8}\mathbb{E}[I^2]. \quad (11)$$

At the same time, the utility obtained by type I under pooling is:

$$\begin{aligned} U_{\text{private}}^{\text{eq}} &= \frac{\mathbb{E}[I]}{2} \cdot v + \left(v - \frac{\gamma \mathbb{E}[I]}{2} \right) \cdot \left(I - \frac{\mathbb{E}[I]}{2} \right) - \frac{\gamma}{2} \left(\frac{\mathbb{E}[I]}{2} \right)^2 \\ &= vI - \frac{\gamma}{8}\mathbb{E}[I](4I - \mathbb{E}[I]). \end{aligned} \quad (12)$$

Computing ex-ante expected utility before inventory realization and using the law of iterated expectations yields

$$\mathbb{E} \left[U_{\text{private}}^{\text{eq}} \right] = v\mathbb{E}[I] - \frac{3\gamma}{8}\mathbb{E}[I]^2, \quad (13)$$

which concludes the proof. $\square$

Proposition 1 implies that privacy is more valuable when inventory risk is higher: that is, either $\text{var}I$ or γ are large. Intuitively, pooling enables risk sharing: high-inventory types—who due to convex costs, have higher marginal liquidation benefits—benefit disproportionately from privacy when dealers cannot discriminate on prices. Therefore, investors purchase privacy ex-ante as it allows them to hedge the risk of receiving large inventories. Two implications follow: (i) If $\text{Var}(I) = 0$, privacy has no value: population heterogeneity is essential for privacy to matter; (ii) Privacy value peaks when H and L types are equally likely. This scenario leads to the greatest dealer uncertainty and thus maximizes the gap between separating and pooling outcomes.

Proposition 2. [Privacy improves welfare] *The ex-ante value of observing the public flow for the monopolistic dealer is $\mathbb{E} \left[\Pi_{\text{public}}^{\text{eq}}(I) \right] - \mathbb{E} \left[\Pi_{\text{private}}^{\text{eq}} \right] = \frac{\gamma}{4} \left(\mathbb{E}[I^2] - \mathbb{E}[I]^2 \right) = \frac{\gamma}{4} \text{var}(I)$, which implies that*

privacy yields a welfare gain of $\frac{\gamma}{8}\text{var}(I)$ since

$$\mathbb{E} \left[U_{\text{private}}^{\text{eq}} + \Pi_{\text{private}}^{\text{eq}} \right] - \mathbb{E} \left[U_{\text{public}}^{\text{eq}} + \Pi_{\text{public}}^{\text{eq}}(I) \right] = \frac{3\gamma}{8}\text{var}(I) - \frac{\gamma}{4}\text{var}(I) = \frac{\gamma}{8}\text{var}(I). \quad (14)$$

Proposition 2 states that privacy raises welfare even though it reduces dealer profits. With public information, the dealer extracts $\frac{\gamma}{4}\text{Var}(I)$ in surplus through targeted price discrimination. Under privacy, that surplus is redistributed to investors, who gain more than the dealer loses because inventory costs are convex. Uniform pricing facilitates cross-type risk sharing: low-inventory investors sell less and high-inventory investors sell more, thereby reducing total inventory holding costs. Consequently, financial privacy acts as a social insurance mechanism: it curbs rent extraction and lowers aggregate risk exposure.

1.4 Equilibrium with reversible privacy

Suppose investors who chose privacy can reveal at $t=1$. A trader reveals only if it improves her price. From (5)–(8), a low-inventory trader prefers revealing whenever $\mathbb{E}[I \mid \text{private}] > L$; a high-inventory trader strictly prefers to remain private since $H > \mathbb{E}[I \mid \text{private}]$.

We consider two informational environments:

- (i) **Externality environment.** In the *externality environment*, the dealer observes sample composition at $t = 1$: n_H high and n_L low inventories. If k_H high-type and k_L low-type investors reveal their endowments, the dealer updates the expected inventory among the remaining private traders as

$$\mathbb{E}[I \mid \text{private, externality}] = \frac{(n_H - k_H)H + (n_L - k_L)L}{n_H + n_L - k_H - k_L}. \quad (15)$$

Hence, each trader's revelation imposes a pricing externality on all others who remain private.

- (ii) **No-externality environment.** In the *no-externality environment*, the dealer does not observe the sample composition and instead treats inventories as independently drawn from the

common prior. The expected inventory among private traders is therefore fixed at

$$\mathbb{E}[I \mid \text{private, no externality}] = \alpha H + (1 - \alpha)L, \quad (16)$$

with $\alpha = \frac{n_H}{n_H + n_L}$. In this case, individual revelation decisions do not affect prices.

Proposition 3. [Revelation with privacy externalities] *Low-inventory traders always reveal at $t = 1$; high-inventory traders never do. That is, $k_L = n_L$ and $k_H = 0$. Therefore, the dealer infers that any private wallets are high-type. The process unravels to full separation, eliminating the ex-ante value of privacy.*

This logic mirrors the standard *unraveling* argument in disclosure models: rational investors fully reveal favorable types, and privacy collapses. In equilibrium, if dealers update prices based on observing public revelation decisions, the willingness to pay for privacy at $t = 0$ is therefore zero, unless behavioral or social frictions impede the unraveling process.

If instead the dealer does not update prices regardless of revelation, the private price remains at $\mathbb{E}[I]$. High types strictly prefer to remain private since pooling gives them a better price than revealing; low types strictly prefer to reveal.

Proposition 4. [Value of privacy with revelation and price commitment] *If the dealer does not update prices and revelation is feasible, the ex-ante value of privacy is*

$$\nu_{\text{option}} = \frac{\gamma}{8} (1 - \alpha) (3H - \mathbb{E}I) (H - \mathbb{E}I), \quad (17)$$

where it follows that $\nu_{\text{option}} > \nu_{\text{binding}}$ for $L > \frac{\mathbb{E}I}{3}$.

2 Experimental design

We implement a laboratory experiment that mirrors the trading game in Section 1. The experiment tests two behavioral frictions that may sustain a positive value of privacy even when revelation is feasible:

- (i) social preferences that internalize the welfare losses disclosure imposes on others, and

(ii) sunk-cost effects that attach additional utility to privacy once paid for.

The design combines between-subject variation in these two dimensions with within-subject variation in inventory realizations to directly link observed behavior to the model's welfare predictions.

2.1 Overview

Subjects act as investors endowed with either low (L) or high (H) token inventory and may sell to a monopolist dealer via a Request-for-Quote (RFQ) interface. Each session replicates the model's decision sequence: privacy acquisition ($t = 0$), inventory revelation ($t = 1$), price quotation ($t = 2 - 3$), and trade execution ($t = 3 - 4$). The key outcomes are the $t = 0$ bids for privacy and the $t = 1$ revelation decisions. To break ties in equilibrium, we assume revelation has a small cost (c_ϵ), which can stand in for gas fees.

We vary two factors between participants: (1) whether privacy is the default or must be actively purchased, and (2) whether privacy externalities are present: That is, whether the dealer price updates with observed disclosures.

Default privacy. In Treatments I and II, investors begin with public wallets and may pay to acquire a private one at $t = 0$ using the [Becker, DeGroot, and Marschak \(1964\)](#) mechanism, before learning their inventory type. This mirrors the theoretical setup in Section 1 where privacy is purchased ex ante as insurance against inventory risk.

In Treatments III and IV, privacy is automatically assigned at the actuarially fair price predicted by Proposition 1. Comparing these treatments isolates the behavioral consequences of actively choosing privacy. If sunk-cost bias matters, participants who pay for privacy will reveal less often at $t = 1$, even when disclosure increases payoffs.

Privacy externalities. The second manipulation concerns whether revelation affects the prices faced by others. In Treatments I and III, the dealer commits to a fixed pooling price based on the population's initial composition:

$$p_{\text{pool}}^{\text{no externality}} = v - \frac{\gamma}{2} (\alpha L + (1 - \alpha) H). \quad (18)$$

Because this price remains constant regardless of disclosures, revelation has no social effect. This condition approximates large, anonymous markets or smart-contract settings in which prices cannot quickly update on realized information. Privacy choices are purely individual: revelation decisions do not generate spillovers to peers.

By contrast, in Treatments II and IV the dealer is Bayesian and revises prices based on the observed share of revealed traders. Whenever low types reveal ($k_L > 0$), the expected inventory among unrevealed traders rises, pushing the price downward. This introduces a negative social externality: individual revelation worsens the terms for others who stay private. This interdependence introduces strategic complementarity in non-disclosure: traders prefer to disclose when they expect others to do the same. However, if social preferences are strong enough, the resulting coordination effect sustains pooling and preserves the ex-ante value of privacy. Table 1 lists the between-participant treatments.

Treatment	Default privacy	Externalities	Revelation possible	Theoretical value
Treatment I	Public	No	Yes	Positive (ν_{option})
Treatment II	Public	Yes	Yes	Zero
Treatment III	Private	No	Yes	Positive (ν_{option})
Treatment IV	Private	Yes	Yes	Zero
Treatment V	Public	No	No	Positive (ν_{binding})
Treatment VI	Public	Fixed price	No	Zero

Table 1: Experimental treatments (between participants)

Treatment V serves as the benchmark with binding privacy: investors begin with public wallets and cannot reveal inventory at $t = 1$, isolating the pure insurance value of privacy without behavioral frictions as given in Proposition 1.

Treatment VI fixes the dealer price to $p = v - \frac{1}{2}\mathbb{E}I$ for both public and private wallets, fully eliminating price discrimination. Any residual willingness to pay in this treatment identifies non-pecuniary (“warm-glow”) value for privacy.

Round structure and timing. Each experimental session consists of five rounds. To become familiar with the platform, all participants start with a short training round (“round 0”). This training round is discarded in the data analysis.

The remaining four rounds implement the core trading game under two variance conditions: two rounds with low inventory variance and two rounds with high inventory variance using the parameters in Table 2. The order of the variance conditions is randomized across participants to eliminate order effects.

Within each variance condition, every participant plays once as a low-inventory (L) trader and once as a high-inventory (H) investor. This within-subject rotation ensures that all participants experience both inventory states under both variance regimes, allowing direct identification of how privacy demand responds to inventory dispersion. In treatments permitting revelation, participants report, before submitting the RFQ, their beliefs about the probability that low- and high-type investors disclose their inventories.

Condition	Inventory		Privacy value		Reveal cost
	High (H)	Low (L)	Binding (ν_{binding})	Option to reveal (ν_{option})	c_{ϵ}
Low	50	70	18.75	46.88	0.10
High	35	85	117.18	152.34	0.10

Note: The asset value is set $v = 20$ and the share of L -types is $\alpha = 0.5$ across all rounds.

Table 2: Within-round parameters and theoretical privacy values.

After the trading rounds, each participant answers a 12-question financial literacy quiz, adapted from [Fernandes, Lynch, and Netemeyer \(2014\)](#) – including the three questions developed by [Lusardi, Mitchell, and Washington \(2011\)](#).

To distinguish between subjective and objective financial literacy, we also ask participants:

On a scale from zero to ten, where zero is not at all knowledgeable about personal finance and ten is very knowledgeable about personal finance, what number would you be on the scale?

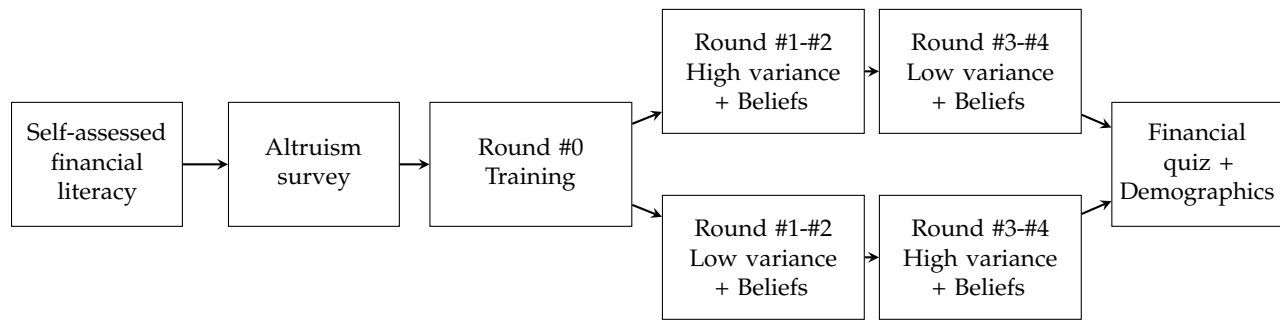
This question measures self-assessed (subjective) financial knowledge and is identical to the one used in [Cupák, Fessler, Hsu, and Paradowski \(2020\)](#). We pose this question before the experiment begins to avoid any influence from the trading game’s monetary performance or the quiz’s perceived difficulty.

Finally, to elicit individual differences in social preferences, we measure altruism using validated items from the [Schwartz \(1992\)](#) value inventory, following the approach of [Brodback, Guenster, and](#)

Mezger (2019) and Brodback, Guenster, Pouget, and Wang (2025). Participants rate the importance of each value statement as “a guiding principle in their life” on an eight-point Likert scale ranging from *Not important at all* to *Of supreme importance*. The resulting scores capture the relative salience of altruistic versus egoistic value orientations, which we use to explain heterogeneity in privacy and revelation behavior across treatments.

Figure 2 illustrates the experiment’s timeline. Appendix ... reproduces the consent form. The experimental instructions given to participants are reproduced in Appendix Before the trading starts, participants need to correctly answer six comprehension questions, listed in Appendix Finally, at the end of the experiment all participants are required to fill in a demographic questionnaire.

Figure 2: Experiment timing



2.2 Hypotheses

From the theory and design, we derive several testable hypotheses:

Hypothesis 1. (Hedging value of privacy) *Bids for private wallets increase with the variance of inventories, $Var(I)$, consistent with privacy acting as insurance against costly inventory shocks.*

Hypothesis 1a. (Financial literacy) *Participants with higher financial literacy exhibit stronger hedging demand for privacy and bid more for private wallets.*

Hypothesis 1b. (Reversibility) *Willingness to pay for privacy is highest in Treatment V (binding privacy) and lowest in Treatments II and IV (revelation with externalities). When revelation unravels privacy, welfare benefits vanish.*

Hypothesis 1c. (Reversibility option) *When revelation is feasible but prices are pre-committed (Treatment I), privacy carries an additional option value relative to binding privacy.*

Hypothesis 2. (Social preferences) *Revelation rates are lower in Treatments II and IV, where disclosure harms others through price updates, than in non-externality treatments (I and III).*

Hypothesis 2a. (Altruism) *Within Treatments II and IV, participants with higher measured altruism reveal less often, mitigating the welfare loss from unraveling.*

Hypothesis 2b. (Belief-driven coordination) *Within Treatments II and IV, revelation rates decline when participants expect fewer others to reveal, reflecting coordination motives in sustaining pooling.*

Hypothesis 3. (Sunk cost effect) *When privacy must be purchased (Treatments I and II), participants reveal less often than in default-private treatments (III and IV), consistent with sunk-cost sensitivity that sustains welfare by limiting unraveling.*

Hypothesis 4. (“Warm glow” non-pecuniary effect) *Willingness to pay for privacy remains positive in Treatment VI, even when privacy yields no pecuniary benefit, indicating intrinsic utility from concealment or autonomy.*

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